Credit Risk Portfolio Analytics

System architecture | Case study

The application combines a multi-source data preparation workflow with portfolio reporting, two classification models, account review tools, and a natural-language interface. The dashboard is implemented in Streamlit and deployed through GitHub to Streamlit Community Cloud.

Data and processing flow

Source data	Preparation	Account-level table	Analysis and models	Dashboard
Customers Loan accounts Bureau credit Repayment history Collections events	Join datasets at account/customer level. Check keys, duplicates, missing values, category consistency, and event dates.	Create demographic, loan, bureau, and repayment aggregates. Define the observed default target and exclude selected leakage-prone fields.	Portfolio segmentation; Logistic Regression and XGBoost; ROC-AUC, KS, Gini, threshold analysis, and SHAP.	Filters and portfolio charts; account risk scoring; anomaly queue; grounded portfolio Q&A.

Application components

Portfolio reporting	Risk scoring	Anomaly review	Ask the Portfolio
Summary metrics and segment-level default rates are calculated from the filtered account data.	Saved Logistic Regression and XGBoost pipelines produce account-level scores. SHAP is used to explain model contributions.	Isolation Forest identifies unusual records. Review status can be tracked and the queue exported.	Deterministic calculations provide supported answers; Gemini is used to phrase responses and must not introduce unsupported portfolio facts.

Deployment: GitHub repository → Streamlit Community Cloud. Gemini credentials are stored in app secrets and are not included in source files.

Limitation: The data does not establish whether all six-month repayment features precede default. Model results are retrospective behavioral classification and are not validated for origination underwriting.